Arunabha Majumder

Design Engineer | Sketch, CAD, Build and Test | Robotics and Wearable Hardware

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SUMMARY

Design engineer who sketches by hand, models in SolidWorks and builds and tests the hardware myself. For my PhD at Aalborg University I took a shoulder exoskeleton from first sketch to a prototype tested with 12 people, cutting energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026). I hold a Diploma in Fine Arts alongside my engineering degrees, so I care about how a product looks and feels to use, not just whether it works.

EXPERIENCE

PhD Researcher, Mechatronics and Robotics (project VIEXO) | 2023 - Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

- Sketched, modelled and built a hybrid shoulder exoskeleton: a spring carries most of the arm load so a small 6 Nm motor is enough, saving more than 25% energy and allowing a smaller, lighter actuator.
- Iterated parts quickly in SolidWorks and 3D printing between test sessions, improving parts that broke or flexed and keeping the CAD up to date with each build.
- Integrated the motor, encoder and force sensors into the frame and built my own test rigs.
- Tested with 12 participants (I collected the data) and worked in a team of 3 on the controller (my part: equations such as the Fourier fit, and how the GUI should behave for the user).
- Validated a force estimate on a 5-bar parallel robot against a reference sensor (within 10% RMS error); presented at international conferences; taught and supervised students (2023 - 2025).

MTech Researcher, Pneumatic Artificial Muscles (Master's thesis) | 2020 - 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated and assembled pneumatic muscles to an existing lab design, collected test data and used it to design the outer PID loop of a gain-scheduled controller (0.3-0.78% error, IEEE CONECCT 2022).

Student Builds, Siddaganga Institute of Technology | 2017 - 2019
Tumakuru, India

- Designed, machined, assembled and user-tested an upper-limb rehabilitation exoskeleton (Best Major Project Award, INR 100,000 TATA Technologies grant); built a Go-Kart and an Efficycle hybrid vehicle.

SKILLS

- **Design:** Hand sketching, concept visualisation, technical illustration (Diploma in Fine Arts)
- **CAD:** SolidWorks (CSWA): parts, assemblies, drawings
- **Build and test:** 3D printing, test rigs; machining and assembly (Bachelor's)
- **Electromechanical:** Motors, encoders, force/torque sensors, DAQ; MATLAB, Python
- **Languages:** English (C1, fluent), Bengali (native), Hindi (conversational)

EDUCATION

PhD, Mechatronics and Robotics (defence expected in 2027) | 2023 - Present
Aalborg University, Denmark

Study abroad, National University of Singapore (NUS), Sep - Nov 2025

MTech, Mechatronics | 2020 - 2022
AcSIR - CSIR-CMERI, India

BTech, Mechanical Engineering | 2015 - 2019
Siddaganga Institute of Technology (VTU), India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. ISRM 2026 (Springer), hybrid actuated shoulder exoskeleton; IEEE CONECCT 2022.

REFERENCES

Available upon request.